

Product Requirement Document (PRD): Loan Simulator

Product Vision

Provide a clear, user-friendly tool that allows users to **calculate their loan payments and view a detailed amortization schedule**.

Benefits of the Project

- **User Empowerment:** Allows users to easily enter loan variables and instantly understand their financial commitment through a clearly calculated fixed monthly payment.
- **Financial Transparency:** Breaks down the repayment process over time by showing exactly how much of each payment goes toward the principal versus interest.

In Scope

- A user interface with a **Loan Calculator capture form** containing required fields: Loan Amount (\$), Annual Interest Rate (%), and Loan Term (months).
- A calculation engine to determine the fixed monthly payment.
- A success notification banner upon successful calculation.
- A detailed **Amortization Schedule data table** that tracks the Period, Principal, Interest, and remaining Balance.
- This is a web application and needs to work in different display resolutions, responsive.

Out of Scope

- **Integration with complex, multiple existing backend systems;** the application is designed to be a small, self-contained solution.
 - **Advanced enterprise loan features** (e.g., variable interest rates, early payoff penalties), as the focus is on a straightforward calculation and payment table.
 - Mobile version is not part of the MVP
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Product Backlog

Epic 1: LOAN DATA CAPTURE

- **User Story 1.1:** Capture Loan Data
- As a user, I want to input my specific loan details so that the system can calculate my unique payment plan.
- **Description:** Implement the user-facing form and API needed to capture the user's loan variables.
 - **Acceptance Criteria:**
 - The main page must display a form titled "**Loan Calculator**".
 - The form must include a mandatory input field for "**Loan Amount (\$)**". Must be in USD currency, the system should put a standard delimited format, most used.
 - The form must include a mandatory input field for "**Annual Interest Rate (%)**". Interest rate must be a numeric value, decimal, and must include keypress restriction.
 - The form must include a mandatory input field for "**Loan Term (months)**". Numeric value, and must include keypress restriction.
 - There must be a primary action button labeled "**Calculate**" at the bottom of the form. The button is required to be inactive by default and should only become clickable once every primary validation for the input fields has been met.

Epic 2: Payment Calculation and Feedback.

- **User Story 2.1:** Display a success calculation message before displaying the loan simulation results.
- As a user, I want visual confirmation that the calculation was successful so I can trust the results on the screen.
- **Description:** The system shall present a toast notification to the user, confirming that the process has been completed successfully.
 - **Acceptance Criteria:**
 - When the calculation is executed without errors, a green banner must appear on the screen.
 - The banner must contain a checkmark and the text "Calculation completed successfully!".
 - Error Handling: If the operation fails, a generic message stating "Something went wrong, please try again" should be shown to cover various execution errors. Furthermore, if a connection issue occurs, the system must display a generic time-out error notification.
- **User Story 2.2:** Generate calculated monthly payment results.

- As a user, I want to see my calculated monthly payment clearly displayed so I know my exact financial obligation.
- **Description:** This is a previous message that appears as a heading of the loan simulator table.
 - **Acceptance Criteria:**
 - Upon clicking "Calculate", the system must process the inputs and display the result under a "Monthly Payment" heading.
 - The monthly payment amount must be prominently displayed (e.g., "\$430.33") with subtext stating "This is your fixed monthly payment amount".
 - Note: This is just an informative heading, not clickable.

Epic 3: Amortization Schedule Generation (Loan Simulation Details)

- **User Story 3.1:** As a user, I want to view a period-by-period breakdown of my loan so I can see my balance decrease over time.
- **Description:** Generate and display a structured breakdown of the loan repayment over its entire lifespan.
 - **Acceptance Criteria:**
 - An "Amortization Schedule" section must be rendered below the monthly payment result.
 - The schedule must be formatted as a data table with the following column headers: "Period", "Principal", "Interest", and "Balance".
 - The table must generate a distinct row for every month defined in the "Loan Term" input (e.g., Periods 1, 2, 3, 4, etc.).
 - The mathematical calculations for Principal, Interest, and Balance must update accurately for each consecutive period. Calculate the formulas on standard financial amortization as example: Calculate an amortization schedule:
 - **Variables:**
 - P = Initial Loan Principal (Total Loan Amount)
 - $AnnualRate$ = Annual Interest Rate (as a percentage)
 - r = Monthly Interest Rate (as a decimal)
 - n = Total Number of Months (Loan Term)
 - M = Fixed Monthly Payment
 - **Formulas:**
 - **Monthly Rate (r):** $(Annual\ Rate / 100) / 12$
 - **Monthly Payment (M):** $\$P * [r(1+r)^n] / [(1+r)^n - 1]$

- **Interest for a specific period:** Previous Period's Remaining Balance * r (For Period 1, the previous balance is the initial loan amount P)
- **Principal for a specific period:** M - Interest for that period
- **New Balance for a specific period:** Previous Period's Remaining Balance - Principal for that period

Examples of Each Variable (From Sources)

Based on the Loan Simulator screenshots in your sources, here is how those generic variables are applied in the application's first period:

- **Initial Loan Principal (P):** \$5000
- **Annual Interest Rate:** 6%
- **Total Number of Months (n):** 12
- **Monthly Rate (r):** 0.005 (Calculated as $6\% / 100 / 12$)
- **Fixed Monthly Payment (M):** \$430.33

Period 1 Breakdown:

- **Interest:** \$25.00 (Calculated as $\$5000 * 0.005$)
- **Principal:** \$405.33 (Calculated as $\$430.33 - \25.00)
- **New Balance:** \$4,594.67 (Calculated as $\$5000 - \405.33)